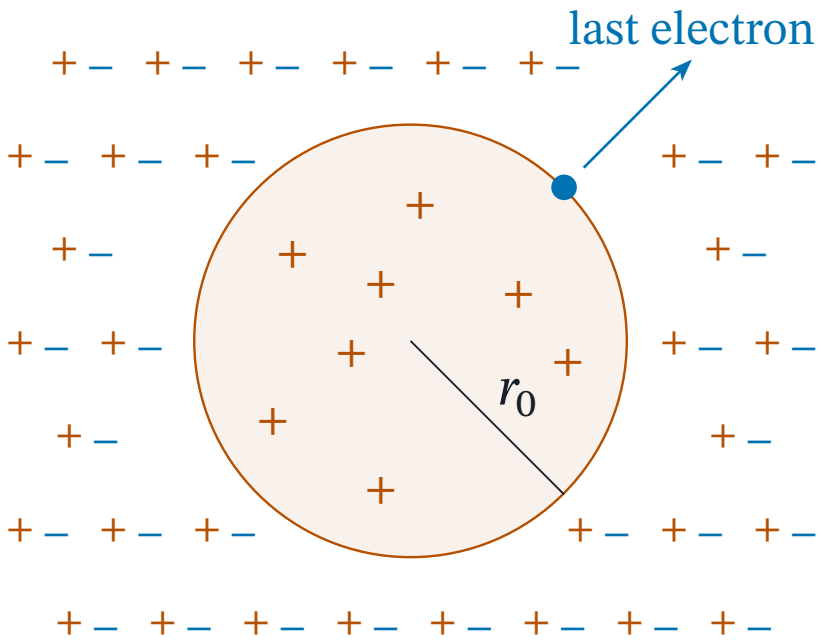
Debye shielding

Christopher Albert

4.1 Charge separation



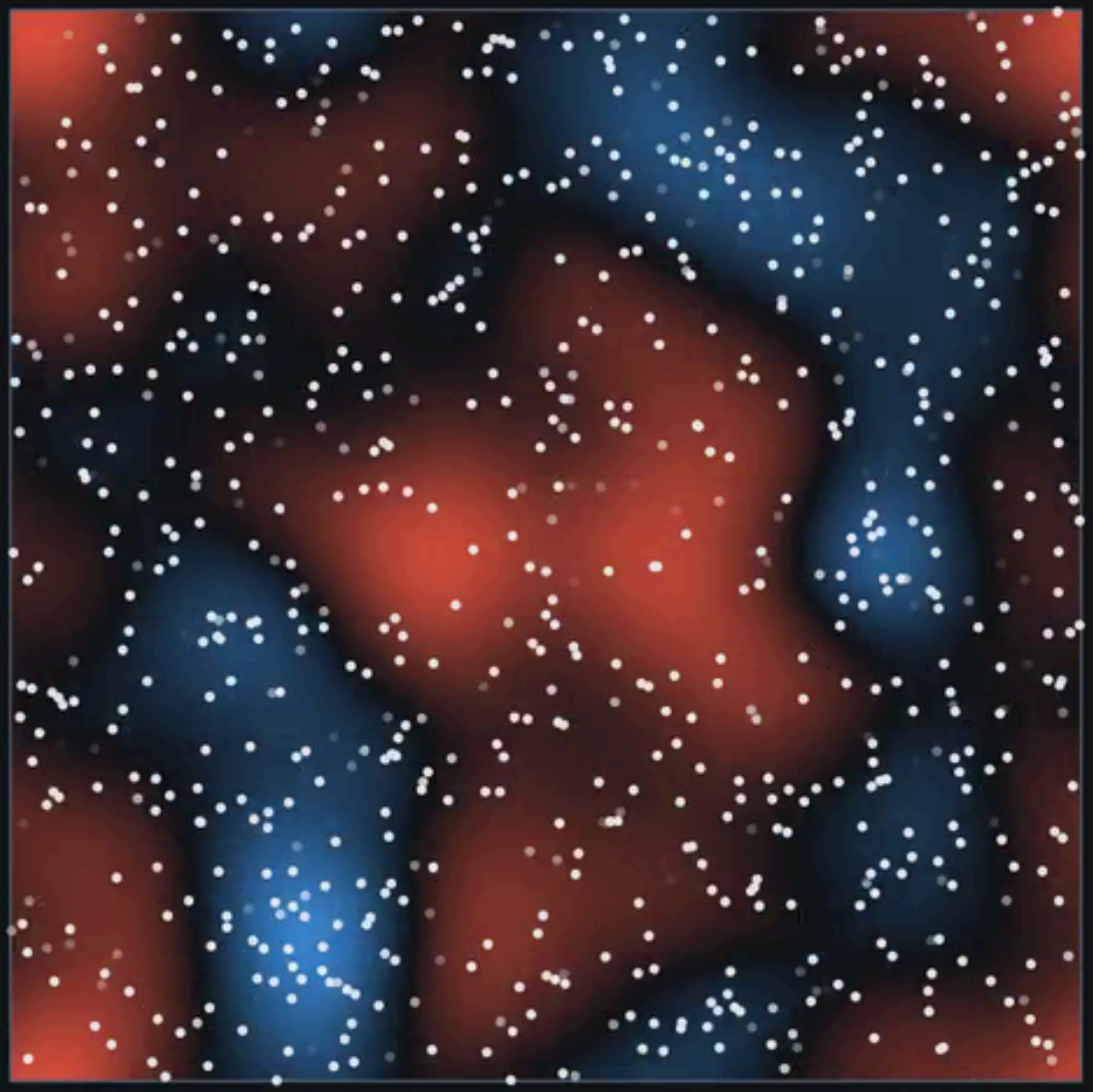
$$Q = \frac{4\pi}{3} r_0^3 n_0 e$$

$$e\Phi(r_0) = \frac{eQ}{4\pi\varepsilon_0 r_0} = \frac{n_0 e^2 r_0^2}{3\varepsilon_0}$$

$$e\Phi(r_0) \approx k_B T_e$$

$$r_0 \approx \sqrt{3}\lambda_D$$

largest separated region



$\rho/(en_0)$ [1]

+0.3

electron deficit
net positive

0

electron excess
net negative

−0.3

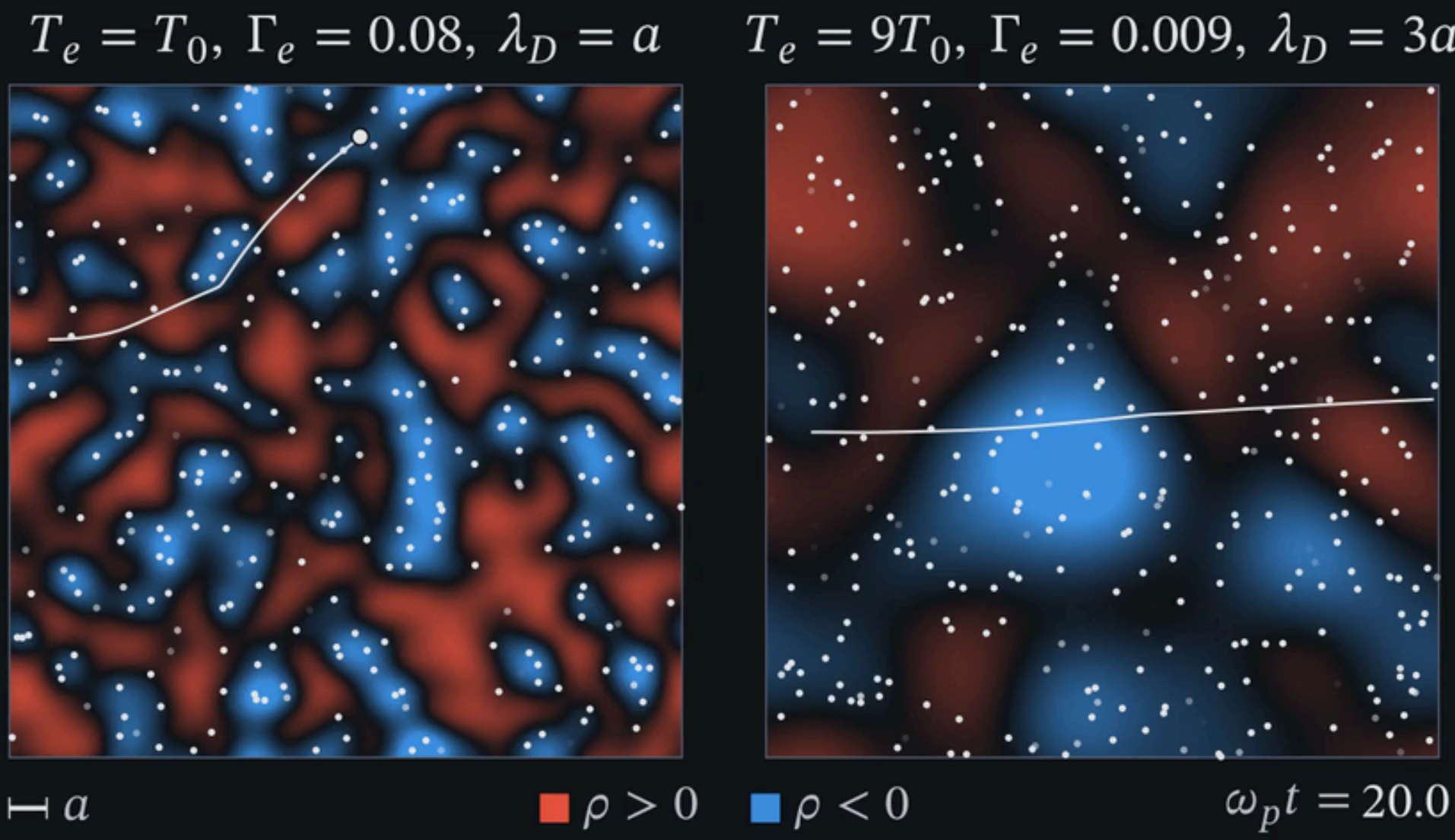
• electron, $|z| < 0.4\,\lambda_D$

smoothing $\sigma = 0.4\,\lambda_D$

λ_D

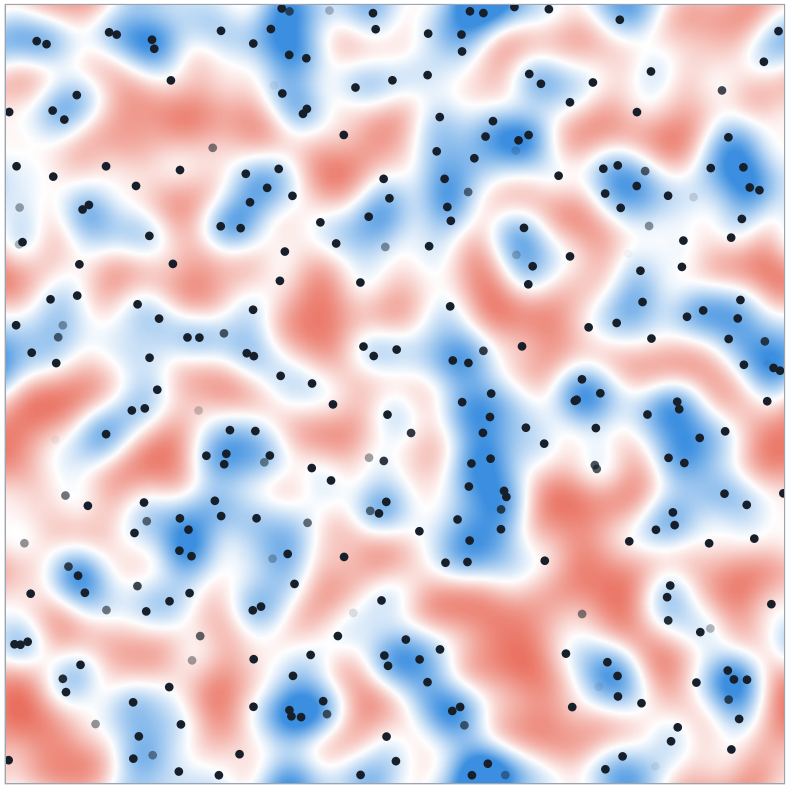
$n\lambda_D^3 = 30$

$\omega_p t = 12.4$


$T_e = T_0,\ \Gamma_e = 0.08,\ \lambda_D = a$
$T_e = 9T_0,\ \Gamma_e = 0.009,\ \lambda_D = 3a$
a
$\rho > 0$
$\rho < 0$
$\omega_p t = 20.0$

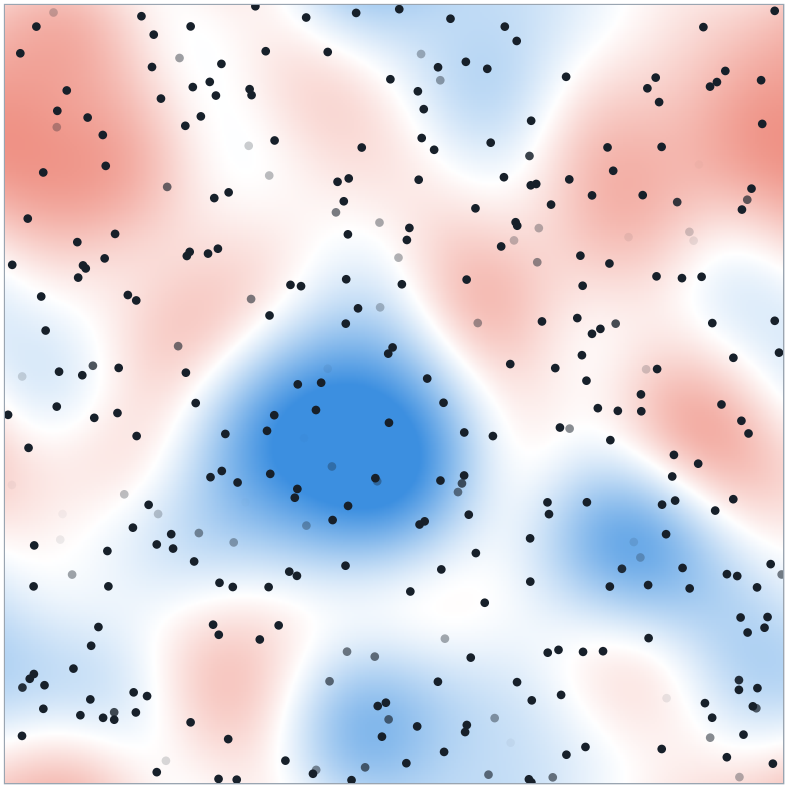
4.1 Weak and strong coupling

$T_e = T_0, \quad \Gamma_e = 0.08, \quad \lambda_D = a$



$\vdash\!\dashv\ a$

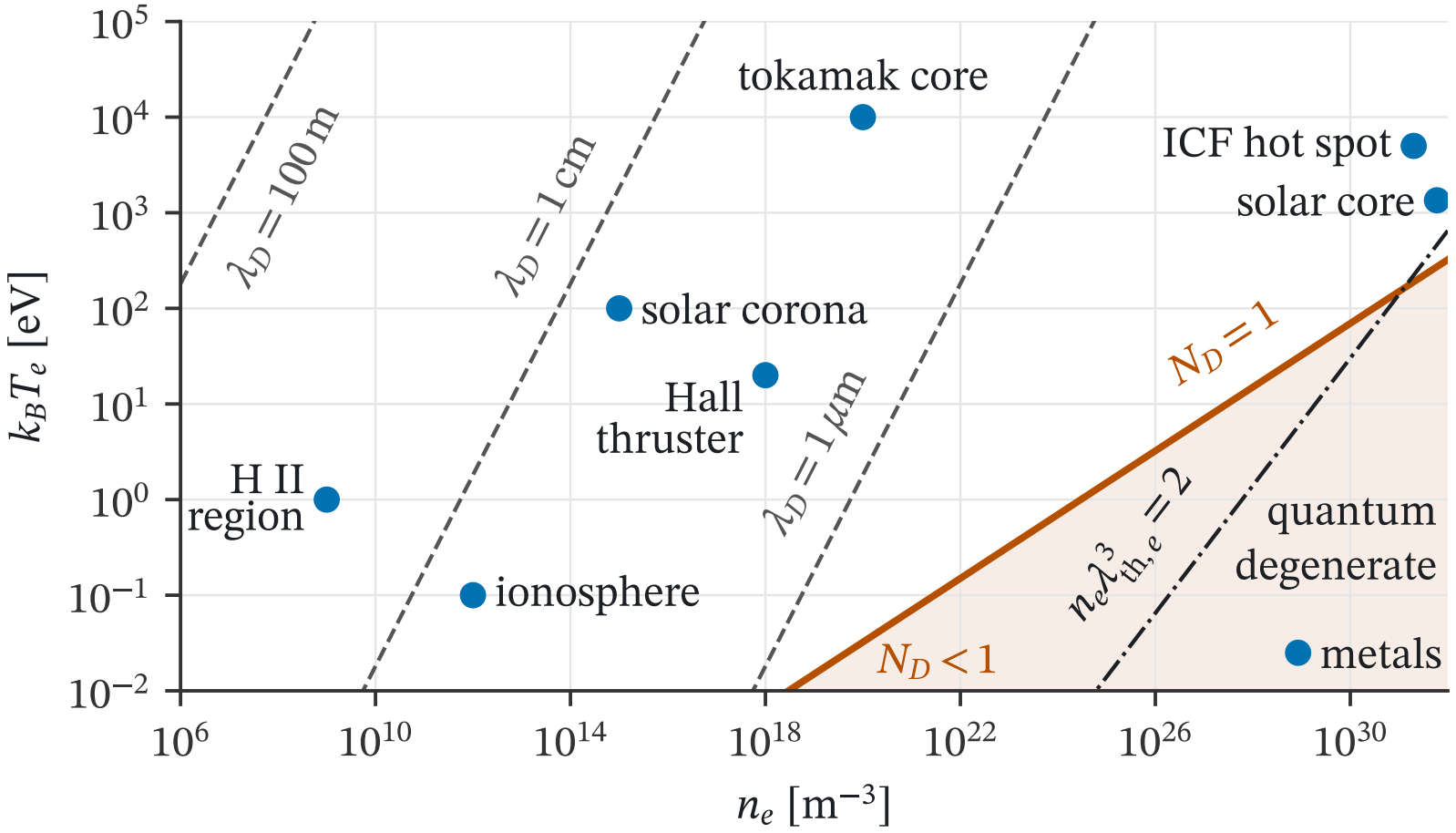
$T_e = 9T_0, \quad \Gamma_e = 0.009, \quad \lambda_D = 3a$



$$\Gamma_e = \frac{e^2/(4\pi\varepsilon_0 a)}{k_B T_e} = \frac{a^2}{4\pi\lambda_D^2}, \quad a = n_0^{-1/3}$$

$$N_D = \frac{4\pi}{3} n_0 \lambda_D^3, \quad \Gamma_e N_D^{2/3} \approx 0.21 \quad \text{weak coupling: } N_D \gg 1$$

4.1 Plasma regime map



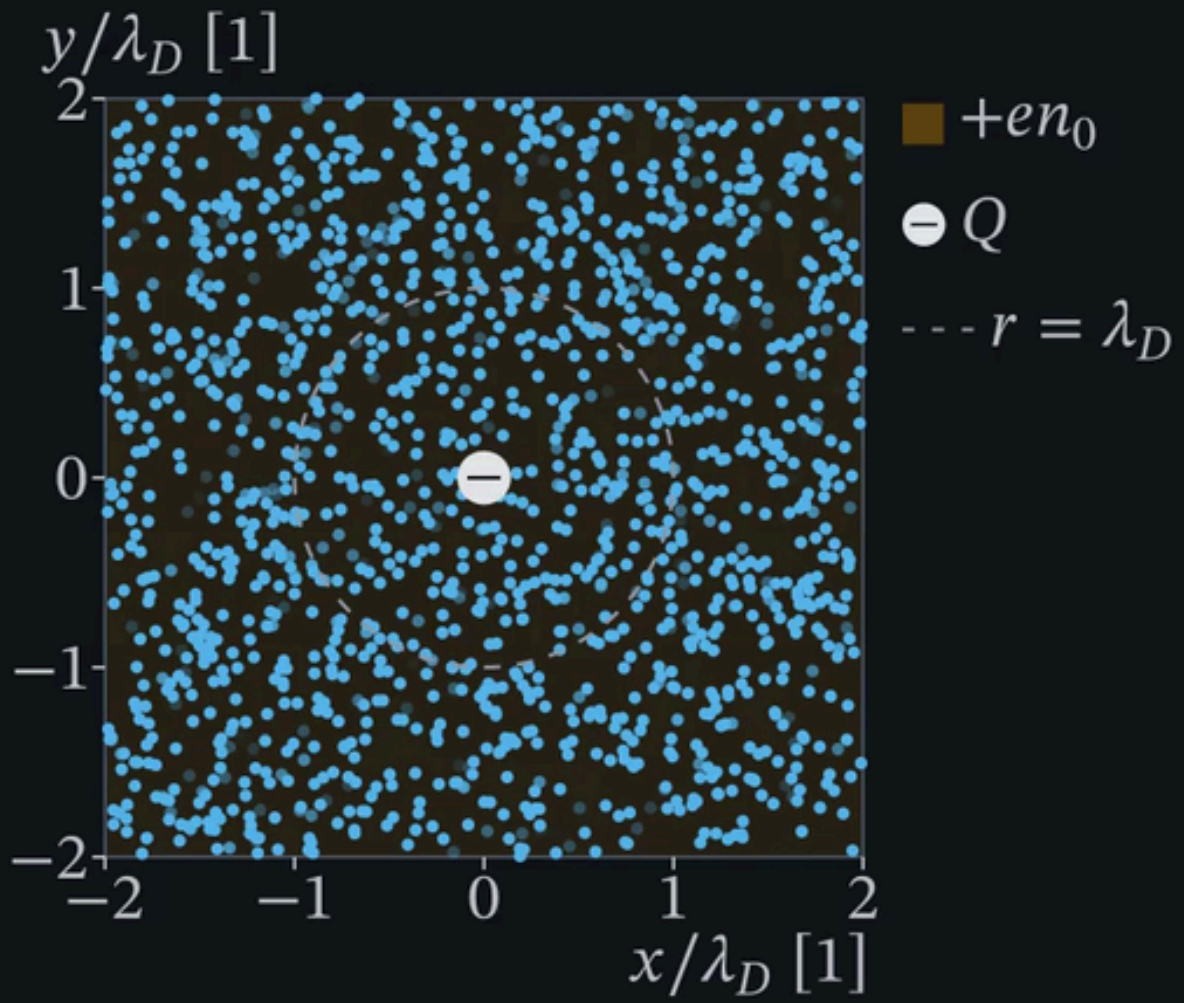
4.1 Ideal plasma

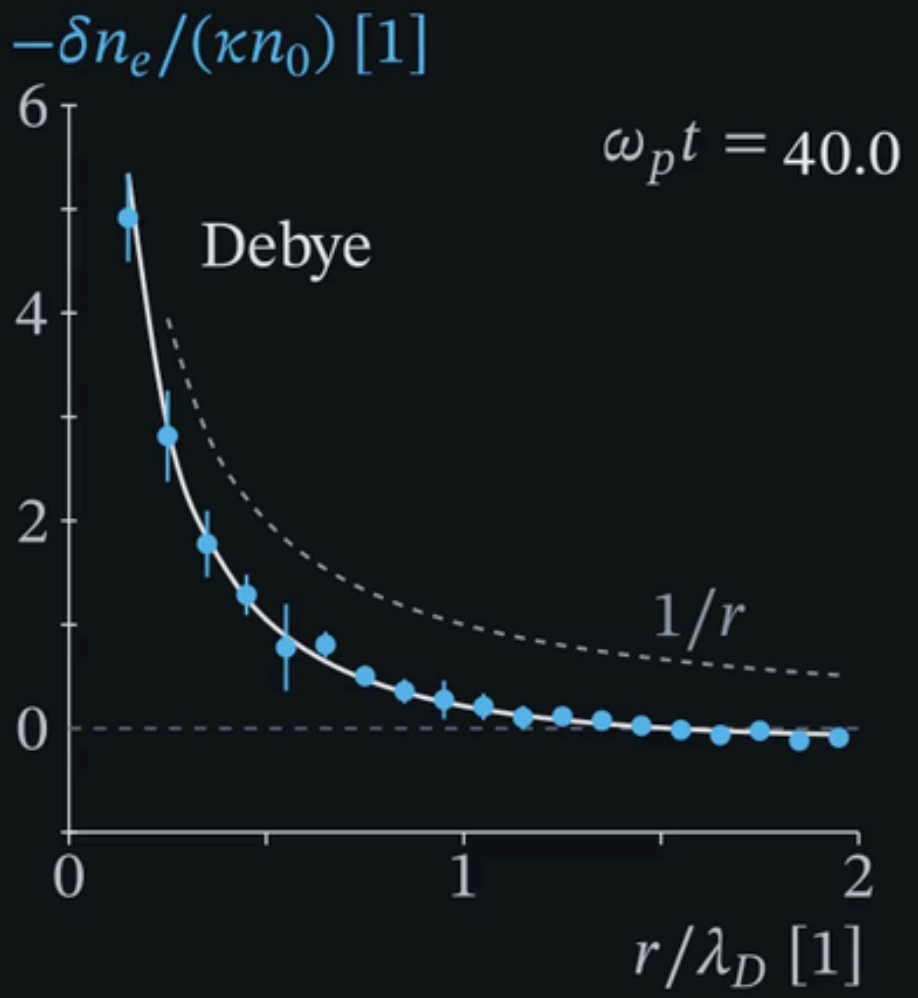
$$N_D \gg 1$$

weak coupling, collective fields

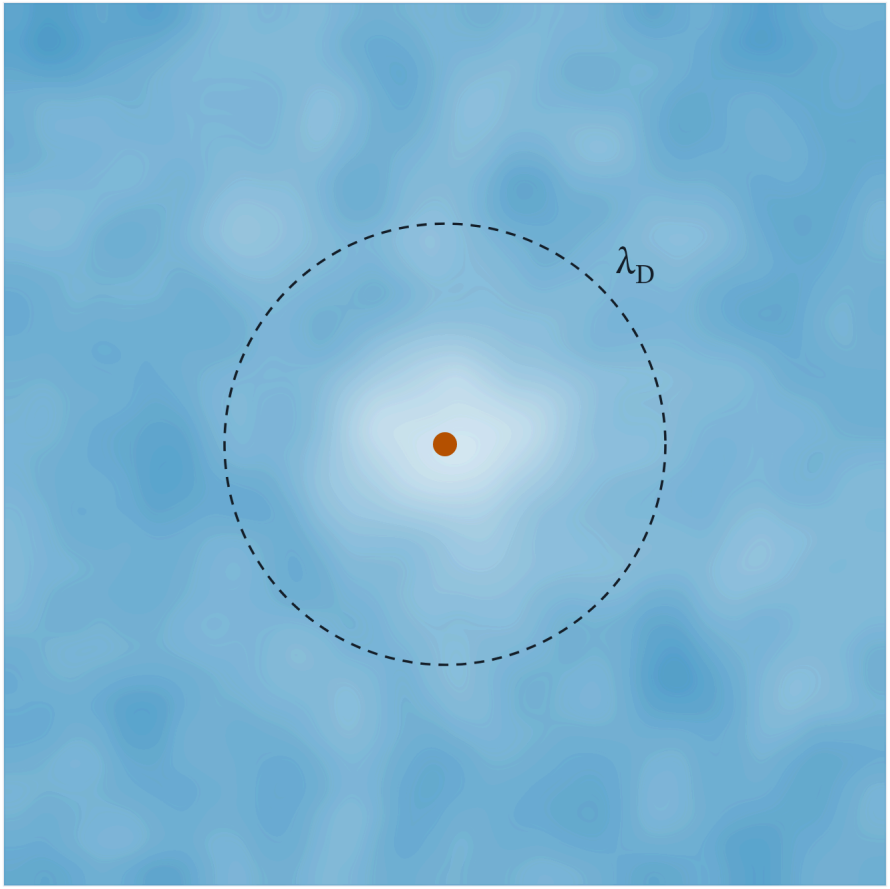
$$\lambda_D \ll L$$

quasineutral bulk; walls affect a few λ_D

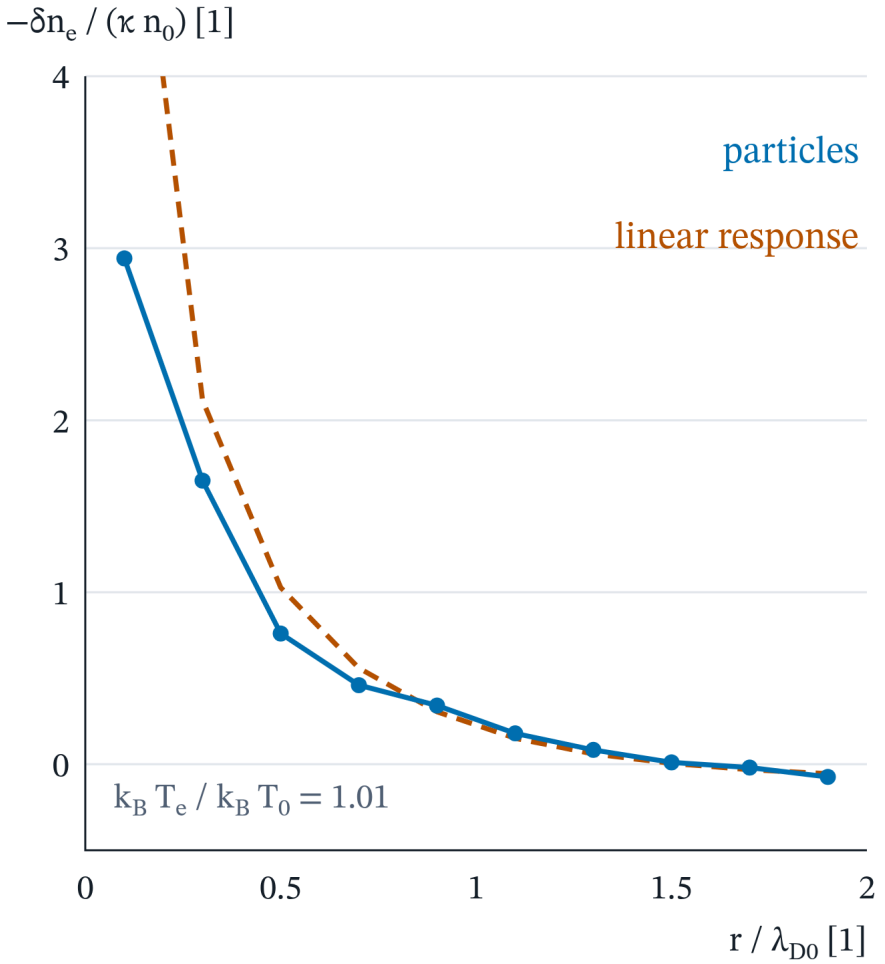

y/λ_D [1]
2
1
0
−1
−2
−2
−1
0
1
2
x/λ_D [1]
$+en_0$
Q
$r = \lambda_D$


$-\delta n_e/(\kappa n_0)$ [1]
6
4
2
0
0
1
2
r/λ_D [1]
$\omega_p t = 40.0$
Debye
$1/r$

4.2 Shielding by particles



$t\,\omega_p = 71$



$k_B T_{bath} / k_B T_0$ 1.00

particles ☐

4.2 Debye shielding

force balance $\quad 0 = -k_B T_e \boldsymbol{\nabla} n_e + e n_e \boldsymbol{\nabla}\Phi \quad \Rightarrow \quad n_e = n_0 e^{e\Phi/k_B T_e}$

Poisson $\quad -\varepsilon_0 \nabla^2 \Phi = e(n_0 - n_e) + Q\delta(\boldsymbol{r})$

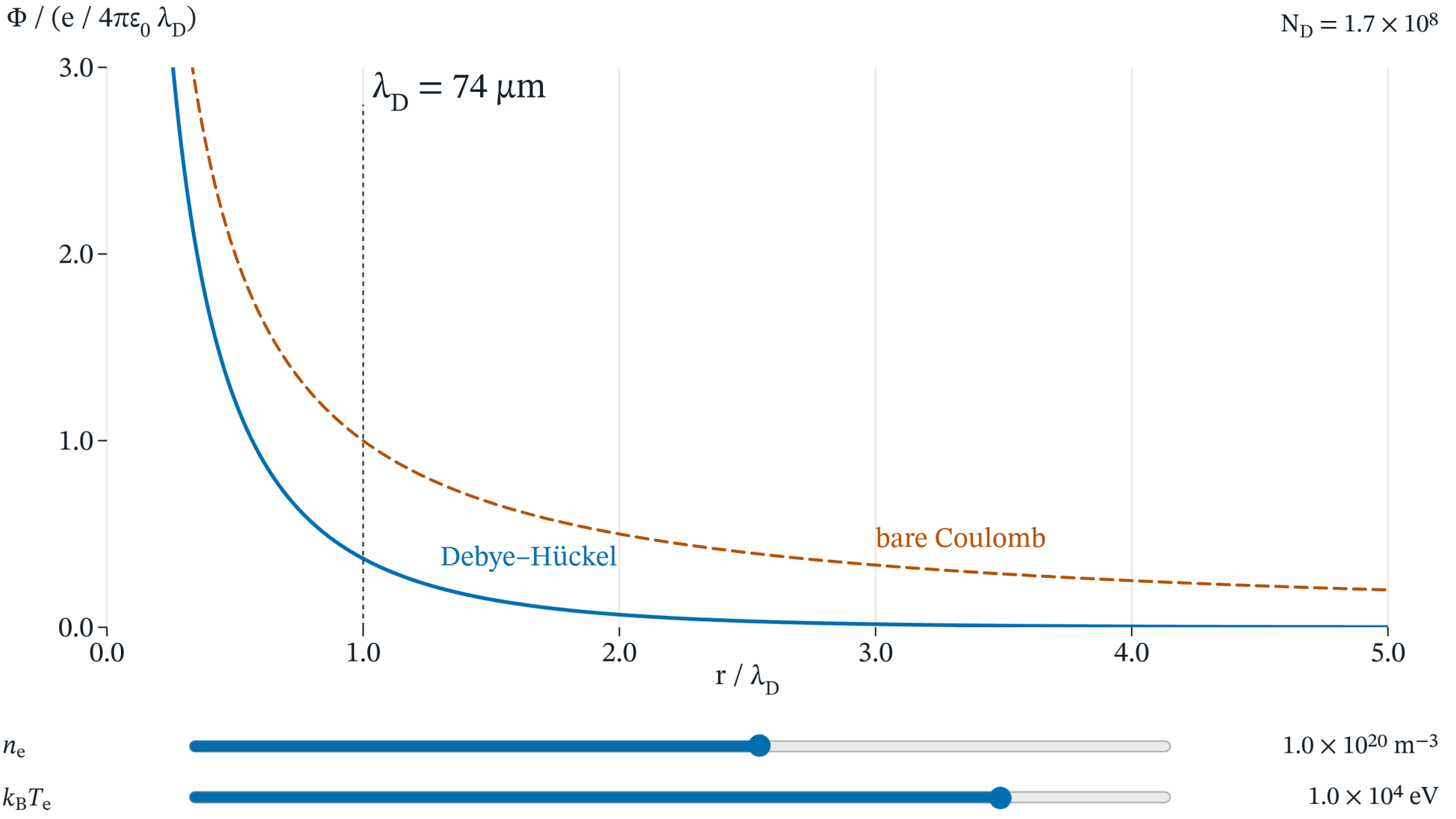
$e\Phi \ll k_B T_e$ $\quad \nabla^2\Phi - \dfrac{\Phi}{\lambda_D^2} = -\dfrac{Q}{\varepsilon_0}\delta(\boldsymbol{r}), \quad \lambda_D = \sqrt{\dfrac{\varepsilon_0 k_B T_e}{n_0 e^2}}$

$\Phi = a(r)/r$ $\quad a'' = \dfrac{a}{\lambda_D^2}$

$$\Phi = \frac{Q}{4\pi\varepsilon_0 r} e^{-r/\lambda_D}$$

Debye–Hückel potential

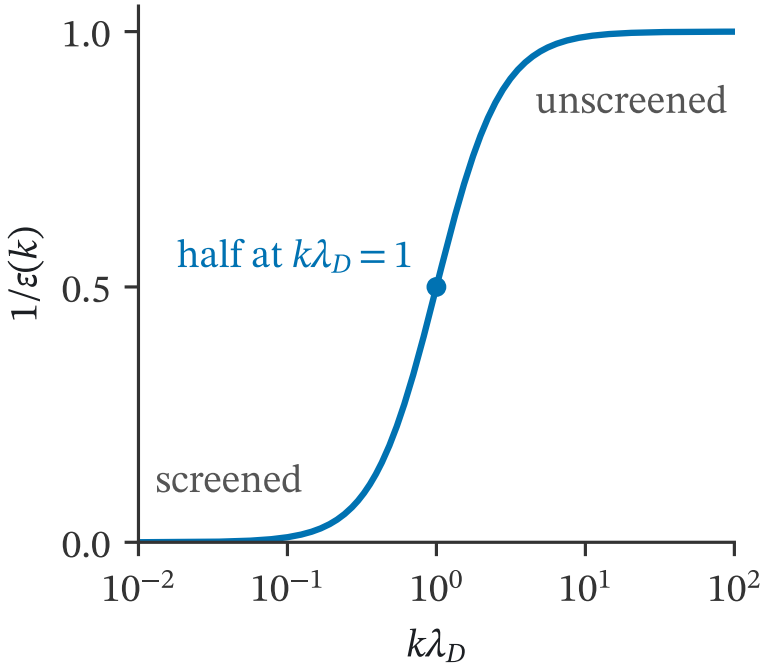
4.2 Screening length



4.2 Dielectric function

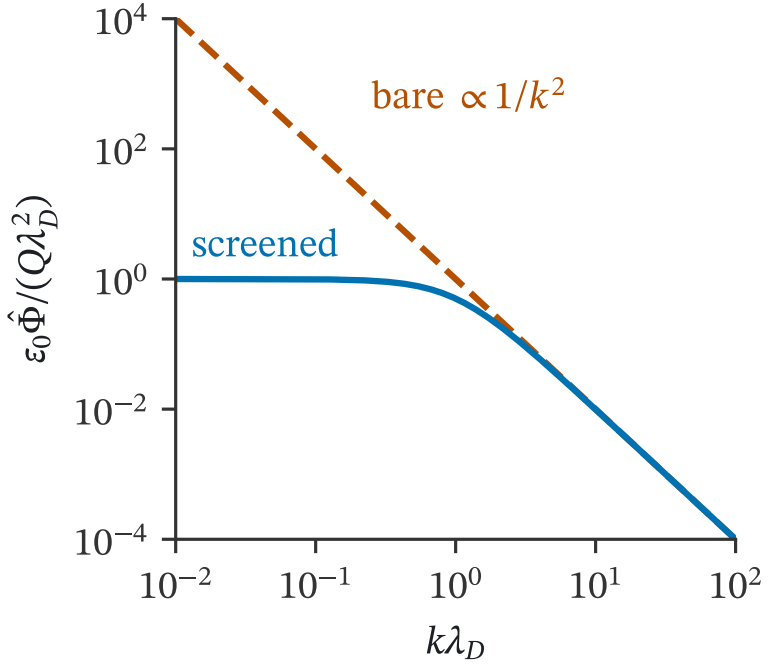
net charge / imposed charge, pattern $e^{i\boldsymbol{k}\cdot\boldsymbol{r}}$

$$\frac{1}{\varepsilon(k)} = \frac{k^2}{k^2 + 1/\lambda_D^2}$$



potential of the test charge $Q\delta(\boldsymbol{r})$

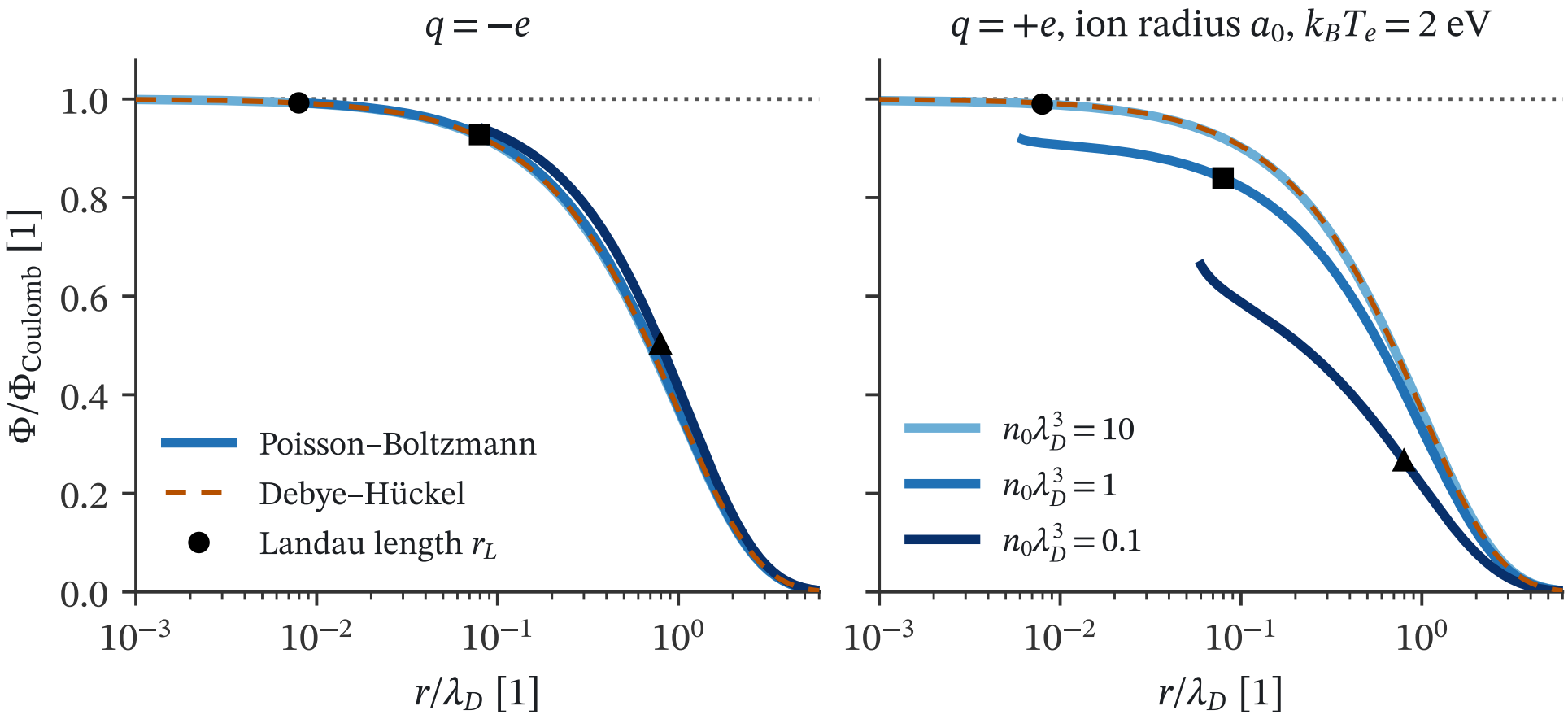
$$\hat{\Phi}(k) = \frac{Q}{\varepsilon_0 k^2 \varepsilon(k)} = \frac{Q}{\varepsilon_0\left(k^2 + 1/\lambda_D^2\right)}$$



$$\varepsilon(k) = 1 + \frac{1}{k^2\lambda_D^2}$$

notebook: Debye shielding

4.2 Nonlinear screening: potential

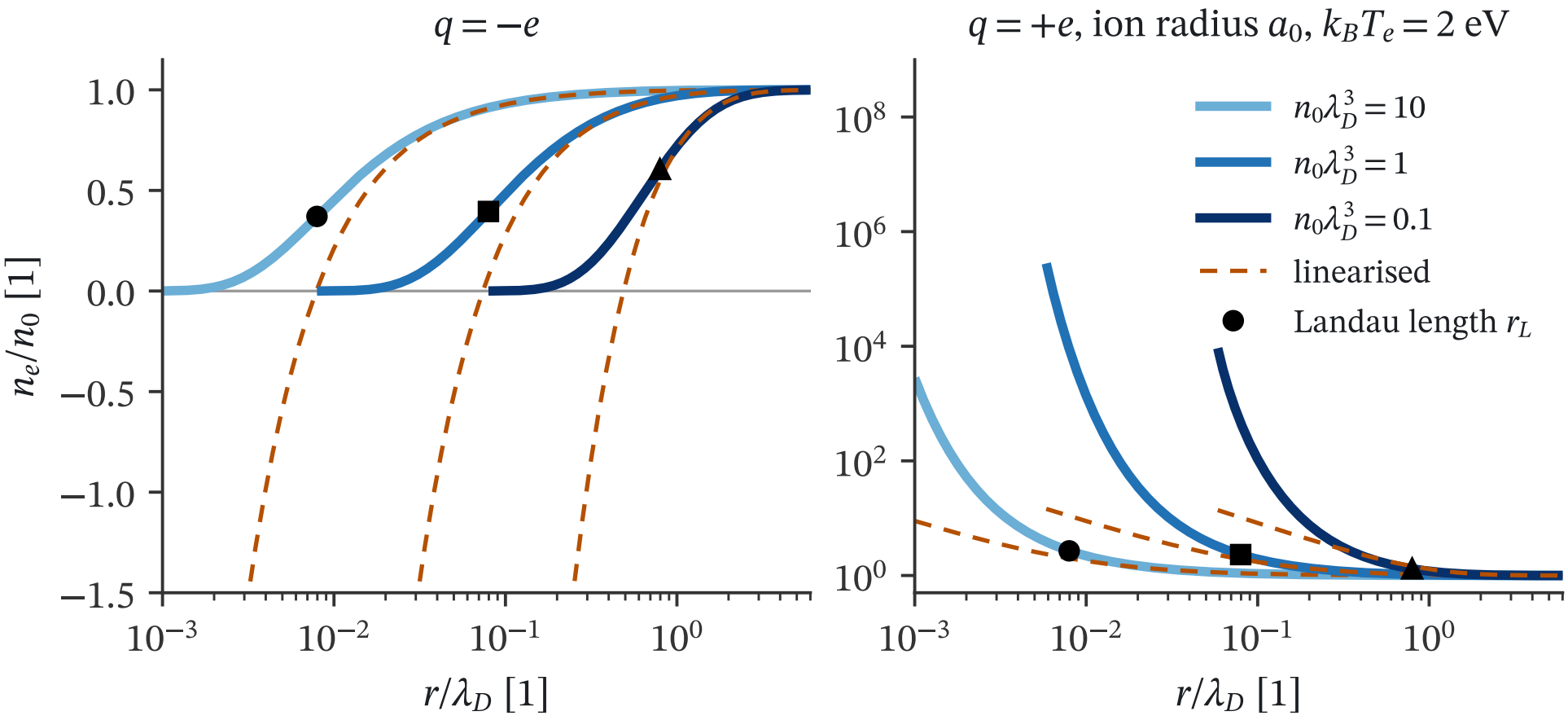


$$r_L = e^2/(4\pi\varepsilon_0 k_B T_e),\ r_L/\lambda_D = 1/\left(4\pi n_0 \lambda_D^3\right)$$

$q = +e$ needs the ion size: $e\Phi(a_0) \approx E_h = 27.2$ eV

visible only for $n_0\lambda_D^3 \lesssim 1$, where mean field itself fails

4.2 Nonlinear screening: density



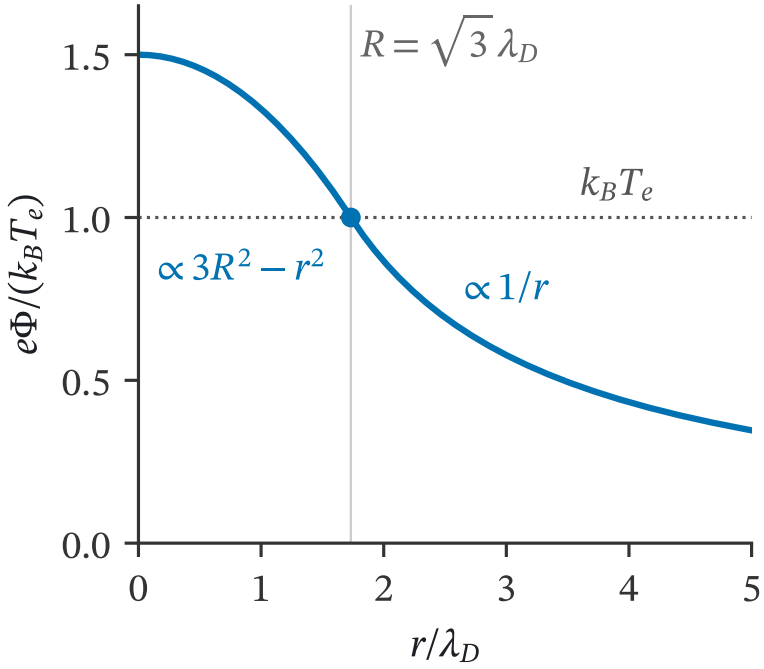
$n_e/n_0 = e^{e\Phi/k_B T_e}$: hole for $q < 0$, pile-up $\sim e^{E_h/k_B T_e}$ at the ion for $q > 0$

up to 0.3 electrons within $2a_0$: a bound state, Saha (chapter 2)

4.2 Charge separation and screening

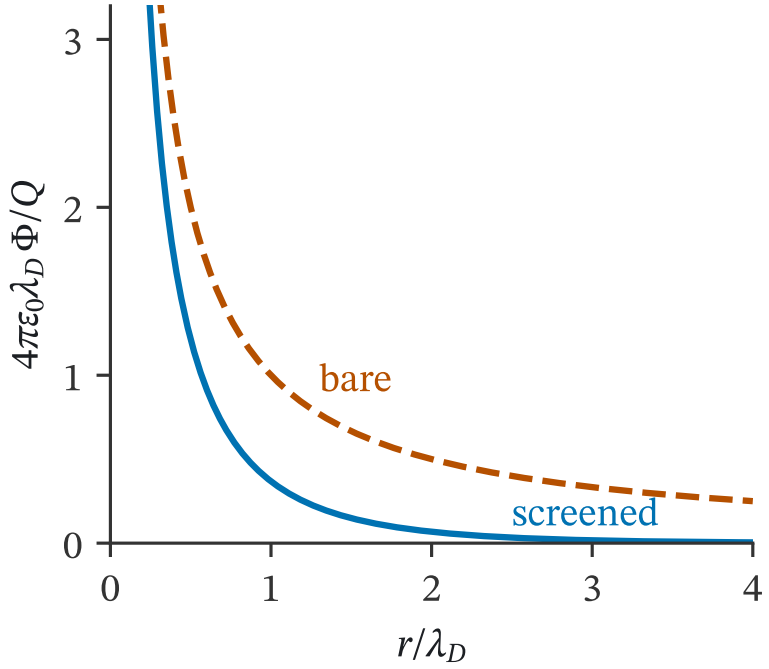
rigid: electrons removed

$$e\Phi(r_0) \approx k_B T_e \quad \Rightarrow \quad r_0 \approx \sqrt{3}\lambda_D$$

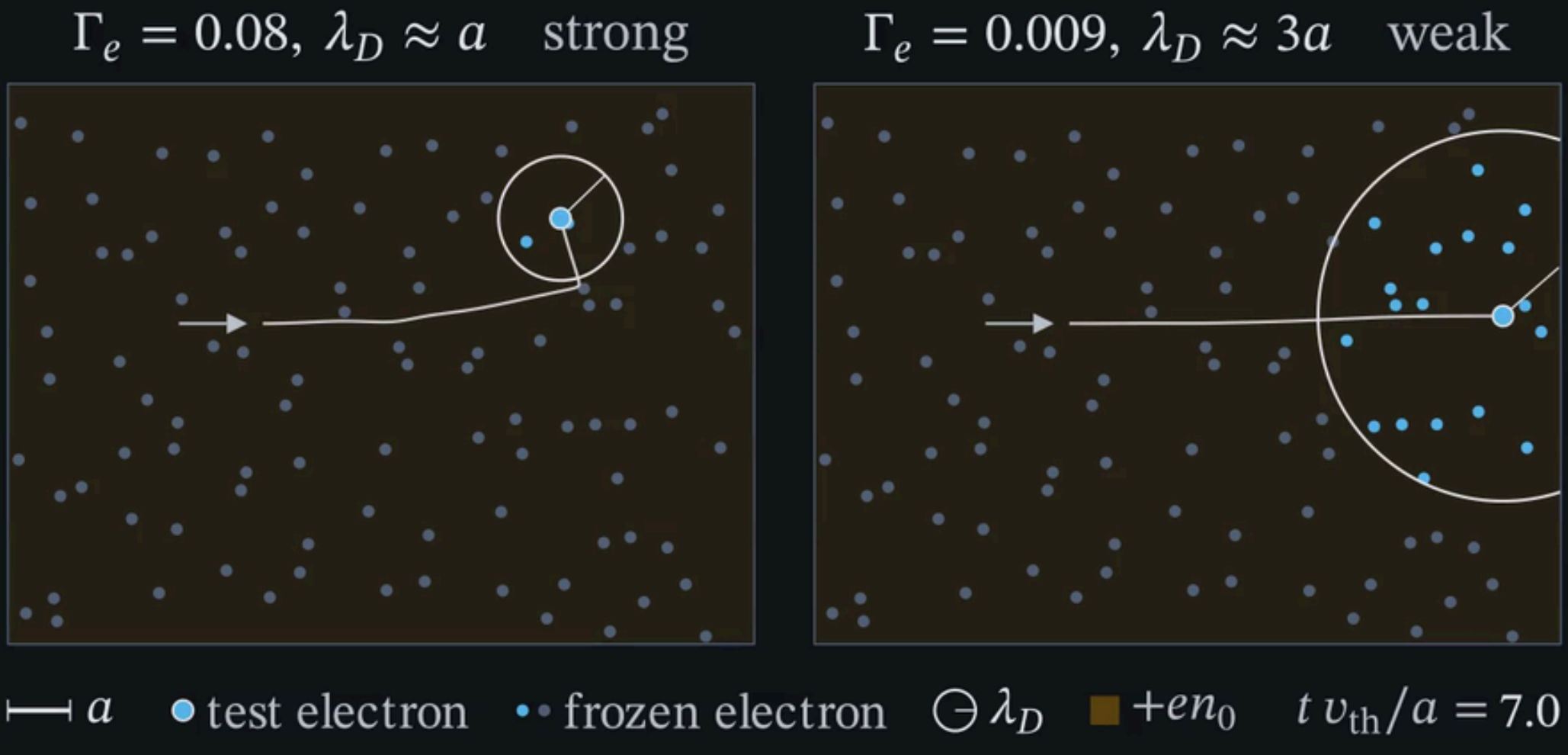


response: mobile electrons

$$\Phi = \frac{Q}{4\pi\varepsilon_0 r} e^{-r/\lambda_D}$$



$$\lambda_D = \sqrt{\frac{\varepsilon_0 k_B T_e}{n_0 e^2}}$$


$\Gamma_e = 0.08,\ \lambda_D \approx a$ strong
$\Gamma_e = 0.009,\ \lambda_D \approx 3a$ weak
a
test electron
frozen electron
λ_D
$+en_0$
$t\, v_{\mathrm{th}}/a = 7.0$

4.2 Typical scales

	n_e [m^{-3}]	$k_B T_e$ [eV]	λ_D [m]	ω_{pe} [s^{-1}]	N_D
H II region	10^{9}	1	2.4×10^{-1}	1.8×10^{6}	5.4×10^{7}
ionosphere	10^{12}	0.1	2.4×10^{-3}	5.6×10^{7}	5.4×10^{4}
solar corona	10^{15}	100	2.4×10^{-3}	1.8×10^{9}	5.4×10^{7}
Hall thruster	10^{18}	20	3.3×10^{-5}	5.6×10^{10}	1.5×10^{5}
tokamak core	10^{20}	10 000	7.4×10^{-5}	5.6×10^{11}	1.7×10^{8}

$$\lambda_D \approx 7400\ \mathrm{m} \sqrt{\frac{k_B T_e / \mathrm{eV}}{n_e / \mathrm{m}^{-3}}}, \qquad \omega_{pe} \approx 56\ \mathrm{s}^{-1} \sqrt{n_e / \mathrm{m}^{-3}}$$

notebook: plasma scales

4.2 Debye length and plasma frequency

$$\lambda_D = \frac{v_{\mathrm{th},e}}{\sqrt{2}\,\omega_{pe}}$$

$v_{\mathrm{th},e} = \sqrt{2k_B T_e / m_e}$

fusion plasma, $n_e = 10^{20}\mathrm{m}^{-3}$, $k_B T_e = 10$ keV:

$$v_{\mathrm{th},e} = 5.9 \times 10^7 \text{ m/s}, \quad \omega_{pe} = 5.6 \times 10^{11}\mathrm{s}^{-1} \quad \Rightarrow \quad \lambda_D = 74\ \mu\mathrm{m}$$